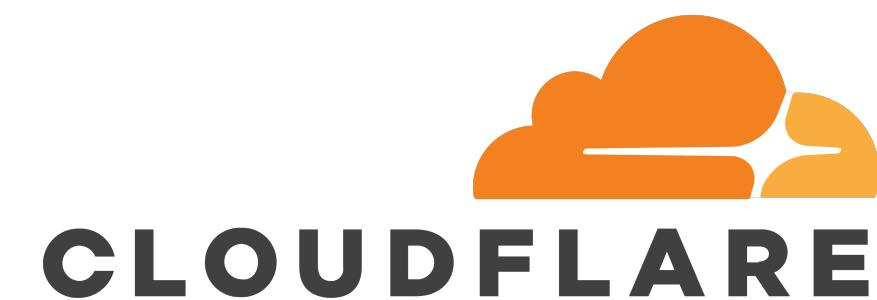




On IP Addresses as Identifiers of Internet Users and Services

Rumaisa Habib, Sudheesh Singanamalla, Marwan Fayed, Zakir Durumeric

Stanford University





This site can't be reached

The connection was reset.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_RESET

Reload

Details

Austrian courts block 11 IP addresses Thousands of domains become collateral



IP blocking has collateral damage

Widely understood but unmeasured

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- We do it so well!

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- CG-NAT
- VPNs, proxies...

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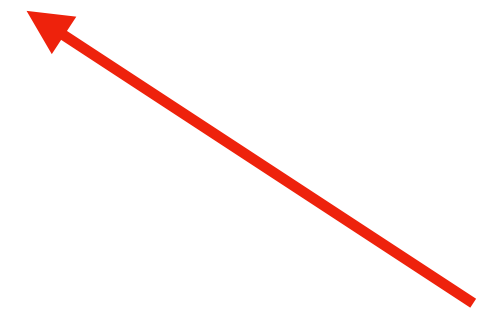
2. Services

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Actually an identifier!

IP addresses were never meant to identify

“It has been considered more useful to deliver the packet, then worry about how to identify the end points, than to provide identity in a packet that cannot be delivered.”

RFC 2101, IPv4 Address Behaviour Today (1997)

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- Cloudflare logs
- 0.1 % of traffic during one day
- 198 countries

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- Active DNS measurements of the top sites

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**The effect changes depending on
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- USA: 5% of IP addresses account for 57% of requests

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None of the obvious explanations hold

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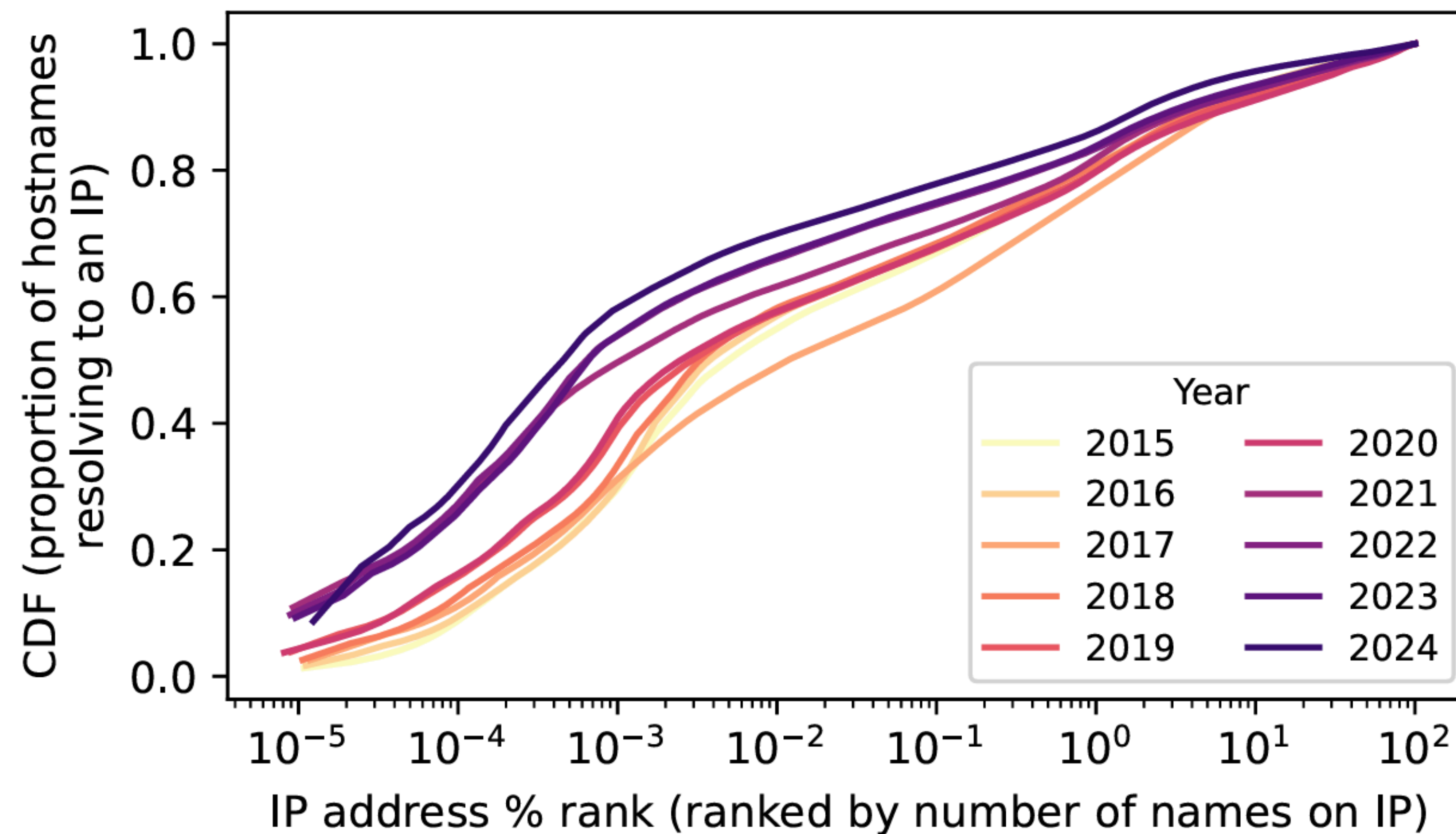
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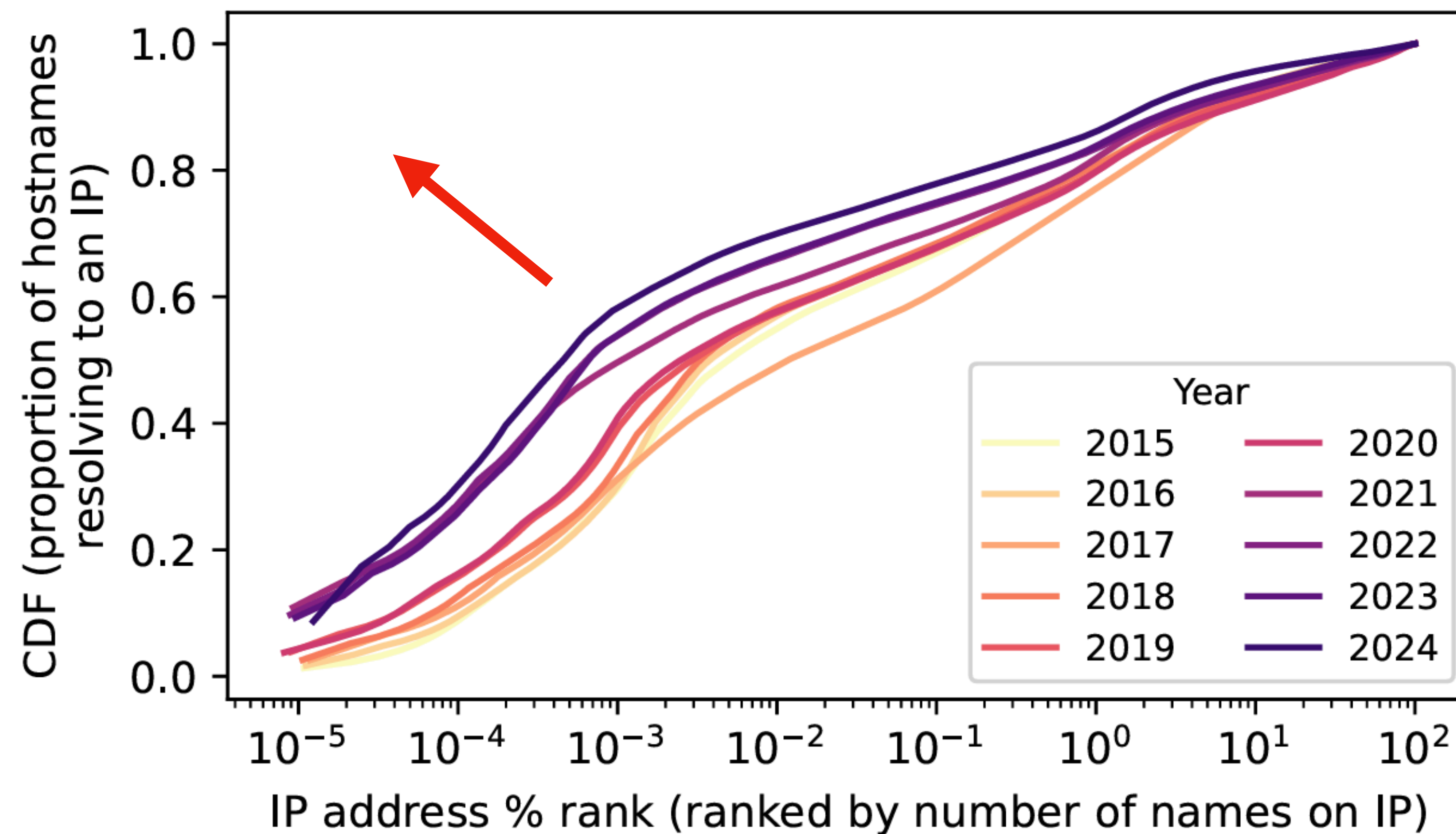
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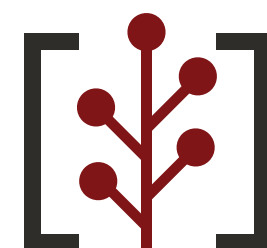
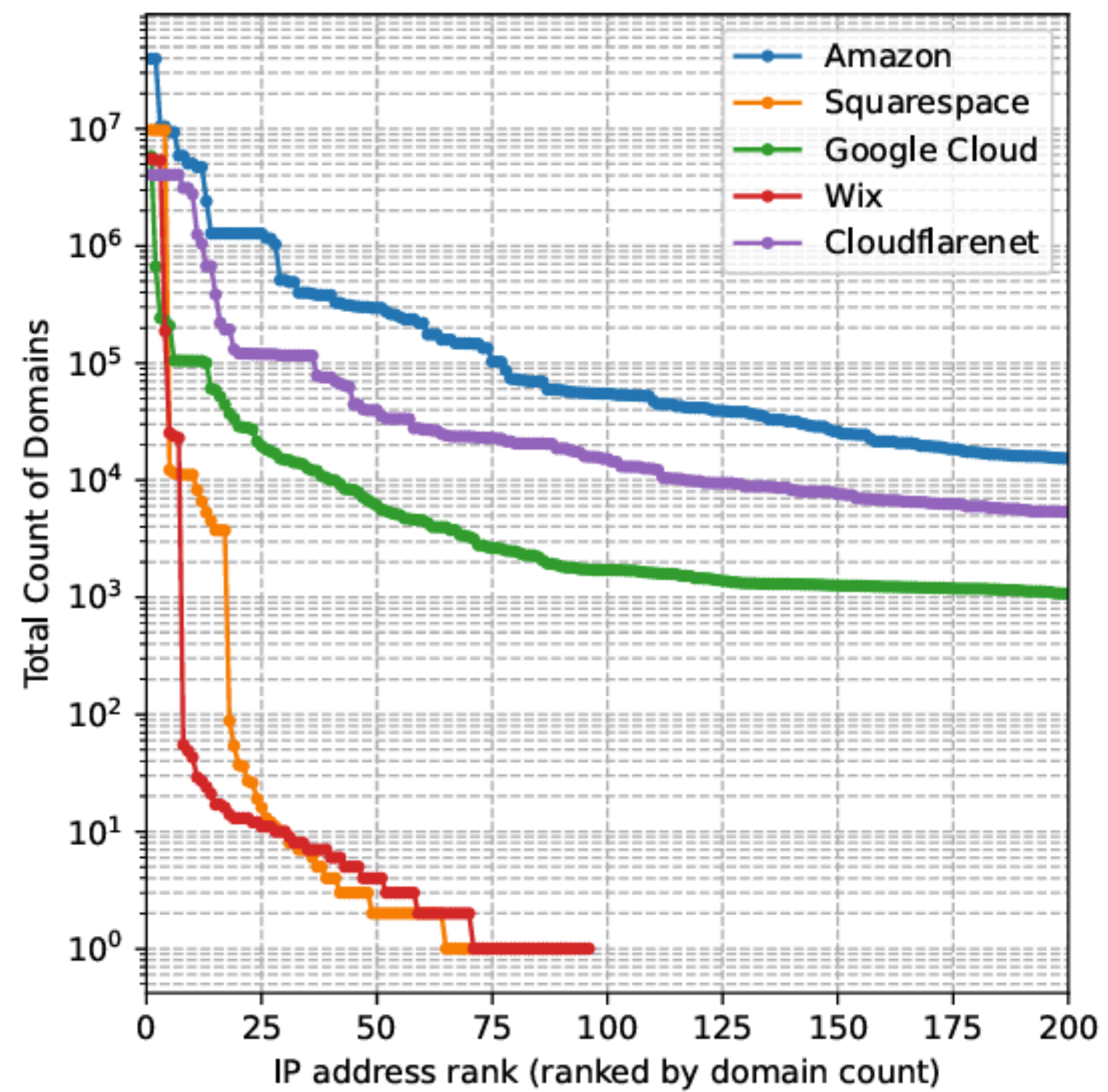
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9% of the top million sites

19% of the top 100 sites







**IP sharing is the norm in the tech industry
Policy hasn't caught up yet**

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Paper 📄

